A Minimal Model of Directional Communication in Honeybee Foraging

Grzegorz Chrupała

May 29, 2026

1 Introduction

Honeybee waggle dances provide spatial information about resources away from the nest. One ecological hypothesis is that such communication is favored when food is difficult to discover independently but remains useful after a scout has found it. Empirical and theoretical studies suggest that the value of dance information depends on the spatiotemporal distribution of food resources, including patch density, patch size, reward variability, and habitat [?, ?, ?, ?, ?, ?, ?]. We begin with a deliberately simple model of directional communication on an exposed horizontal comb, where dance angle can map directly onto food direction.

2 Method

Colonies are the reproducing entities, while workers are behavioral agents. Each colony has four heritable mean traits: directional bias, which controls how strongly a scout's dance angle is centered on the intended direction; receiver attention, which controls how likely recruits are to follow the dance; sender transposition, which controls how much the scout maps food direction onto a gravity-referenced dance angle; and receiver transposition, which controls how much recruits interpret the dance as gravity-referenced rather than direct pointing. Individual workers are sampled around the colony means using stable within-colony variation. Each dance also includes transient production and interpretation noise.

More concretely, a colony is parameterized by mean directional bias $b \in [0, 1]$, mean receiver attention $a \in [0, 1]$, mean sender transposition $s \in [0, 1]$, and mean receiver transposition $r \in [0, 1]$. Each worker receives stable individual deviations from these means, sampled from Gaussian noise with standard deviation 0.08 and then clipped to the unit interval. These deviations represent within-colony worker variation, but are not themselves heritable in the current model. At reproduction, only the colony-level mean traits are inherited, with Gaussian mutation of standard deviation 0.07.

Each foraging episode generates a set of food sites in the world. A food site has a direction, angular width, value, and capacity. A scout discovers one food site and dances for it. Recruits either attend to the dance and search near the signaled direction, or ignore it and search in a random direction. Crucially, independent searchers can find any available food site, not only the scout's site. This makes background discovery an emergent property of the food distribution. This design follows the logic of previous ecological models in which recruitment benefits depend on the availability and discoverability of resource patches rather than on communication having a fixed payoff advantage [?, ?].

Comb tilt is represented by an environmental parameter $t \in [0, 1]$, where $t = 0$ is a horizontal comb and $t = 1$ is a vertical comb. Direct pointing has quality $1 - t$: it is reliable on a horizontal comb and uninformative on a vertical comb. Gravity-referenced mapping remains available at all tilts. The sender transposition trait blends continuously between direct pointing and gravity-referenced encoding, and the receiver transposition trait blends between direct and gravity-referenced decoding. The initial food-distribution experiments below use $t = 0$, so the setup reduces to the horizontal-comb case.

The scout's dance angle is sampled from a circular von Mises distribution centered on the encoded direction. The concentration parameter is $14b$, so workers with low directional bias produce nearly random angles, while workers with high directional bias produce angles concentrated around the intended direction. Additional dance-production noise with standard deviation 0.18 is added to the angle. A recruit attends to the dance with probability a; if it attends, it decodes the signaled direction with interpretation noise of standard deviation 0.12. If it does not attend, it searches in a uniformly random direction. A recruit succeeds when its search direction falls within the angular width of any food site that still has remaining capacity.

After many episodes, each colony receives a payoff from collected food minus communication and attention costs. Daughter colonies are sampled in proportion to payoff, and colony-level traits mutate by Gaussian noise. We track the mean directional bias, receiver attention, sender and receiver transposition, and recruit success rate across generations.

In each episode, collected food is the sum of the values of food sites found by recruits. The current simulations use food value 1.0. The colony then pays a communication cost of $0.02b_s$, where b_s is the scout worker's directional bias, and an attention cost of 0.01 for each recruit that attends to the dance. Thus more directional signaling and more dance-following can increase food collection, but they are not free. The current model does not include an independent colony maintenance cost. The parameter "workers per colony" controls the pool from which scouts and recruits are sampled, and therefore the amount of within-colony worker heterogeneity, but payoff opportunity is set more directly by the number of foraging episodes per colony and the number of recruits per episode.

2.1 Spatially explicit extension

We also implemented a separate spatially explicit prototype to verify that the core evolutionary effects are robust when resource discovery depends on location and travel. In this extension, food patches are placed as circular sites in a two-dimensional square world. Scouts communicate a noisy target coordinate for a discovered patch, and recruits travel from the hive to the signaled point. Travel cost is proportional to Euclidean distance, and recruits that do not attend to the dance instead select a uniformly random target point. A recruit succeeds when it arrives within a patch that still has remaining capacity. This spatial prototype retains the colony-level reproduction and mutation structure of the original model while adding world size, patch radius, forager speed, travel cost, perception radius, and coordinate signalling noise as explicit parameters.

3 Results

3.1 Horizontal comb

We compared three simple food distributions over ten random seeds each. The "hard" condition had one narrow food site per episode, the baseline condition had two moderately wide food sites, and the "easy" condition had many broad food sites. Table ?? reports the generation at which mean directional bias first reached 0.30, plus final generation trait and success values.

Each run used 60 colonies, 80 workers per colony, 30 generations, 50 foraging episodes per colony per generation, 6 recruits per episode, and horizontal-comb tilt $t = 0$. Food-site value was 1.0, and all three conditions used food-site capacity 6. The conditions varied only the number and width of food sites: hard used one site of width 0.08, baseline used two sites of width 0.20, and easy used eight sites of width 0.50. The hard condition therefore makes independent discovery unlikely, while the easy condition makes independent discovery likely even without using the dance.

Condition	Reach generation	Final bias	Final attention	Final success
Hard	11.0	0.741	0.871	0.144
Baseline	11.8	0.745	0.826	0.373
Easy	23.9	0.353	0.455	0.811

Table 1: Initial food-distribution experiment. The hard environment used one food site of width 0.08, the baseline used two sites of width 0.20, and the easy environment used eight sites of width 0.50. All conditions used food-site capacity 6 and ten seeds.

The initial result matches the ecological prediction qualitatively. In the hard environment, communication evolves quickly and strongly, although absolute foraging success remains low because food is rare and narrow. In the easy environment, recruit success is high even without strong communication, and selection for directional bias and receiver attention is much weaker. The baseline condition lies between these extremes.

3.2 Spatial prototype comparison

We also compared a spatially explicit prototype using parameters that are comparable to the original horizontal-comb experiment: 60 colonies, 80 workers per colony, 30 generations, 50 episodes per colony, 6 recruits per episode, and the same communication costs and patch values. The spatial model used a 100-by-100 world, moderate travel cost, and noisy coordinate signalling. Under these comparable settings, the spatial prototype also showed evolution of directional signaling and receiver attention, with mean success rising from about 0.28 to about 0.93 over 30 generations. This suggests that the basic selection dynamics survive the move to explicit space, even though the mechanistic interpretation of the signal and the underlying parameters are different.

3.3 Vertical comb

We also made a first exploratory comparison of horizontal, tilted, and vertical combs while holding the baseline food distribution fixed. In this run, comb tilt was set to $t = 0$, $t = 0.5$, or $t = 1.0$, and each condition used a single seed. Table ?? summarizes the result after 30 generations.

As expected, direct horizontal pointing performs best when the comb is horizontal. Increasing comb tilt weakens the existing direct mapping. In the vertical condition, sender transposition rises substantially, indicating selection toward gravity-referenced encoding, but receiver transposition and overall communication success remain limited after 30 generations. A longer single-seed run to 80 generations suggested that sender and receiver transposition can coevolve further, but this needs a systematic experiment.

Comb	Final bias	Final attention	Sender trans.	Receiver trans.	Final success
Horizontal	0.835	0.886	0.318	0.238	0.412
Tilted	0.691	0.713	0.385	0.333	0.189
Vertical	0.278	0.292	0.753	0.473	0.143

Table 2: Exploratory comb-tilt comparison using the baseline food distribution and one seed. "Sender trans." and "Receiver trans." denote the final mean sender and receiver transposition traits.

4 Conclusion

Placeholder.